

1) city hospital data warehouse.

a) star schema.

Fact table: consultation - Fact.

* measures: count, charge, avg-charge.

Dimension tables:

* patient - dim (id, name, Age, Gender).

* doctor - dim (id, name, specialization).

* Department - dim (id, name).

* time - dim (day, month, quarter, year).

b) OLAP operations:

* Roll-up: time $\rightarrow$ year.

* Slice: select Average charge.

* Dice: department-wise data.

c) no. of cuboids:

formula:

$$(n+1)^d$$

$$= (4+1)^4 = 625 \text{ cuboids.}$$

2) smart city project:

star schema

Fact Table: smartcity - Fact.

measures:

* vehicle count.

* congestion Level.

- * punctuality.
- * Air quality index (AQI).

Dimensions:

- * time.
- * location.
- * transport
- * sensor.

3.) Banking data warehouse:

a.) Galaxy schema.

Fact tables:

- * Transaction - Fact. (Amount, count).
- * Loan - Fact. (Amount, interest).

shared dimensions:

- * customer.
- * Branch
- * time.
- * Account type.

b.) OLAP operations

- * Roll-up: time $\rightarrow$ year.
- * Dice: branch-wise data.

measure: Average transaction Amount

Loan Amount.

c.) cuboids:

$$(4+1)^4 = 625 \text{ cuboids.}$$

Manufacturing company:

a) Roll-up
* summarize defect rate by product category across all factories for the current year.

b) Drill-down
* Analyze defect rate by factory → department → product type.

c) slice.

select suppliers with rating = 4 + across all regions.

d) dice

* Analyze defect rate and supplier reliability for selected product categories & factory locations.

5.) Retail chain.

a) Roll-up.

* summarize inventory turnover by product category across all stores for the last quarter.

b) Drill-down.

* Analyze electronics sales: Region → category → SKU.

c) slice.

* view total sales revenue for male customers with bronze membership.

d) dice.

* Analyze sales revenue and units sold for selected cities, product categories and time periods.

